USER MANUAL

Student Grade Management System

A comprehensive tool for instructors to manage, analyze, and visualize student grade data

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Designed to support instructors in managing class performance efficiently, accurately, and with actionable insights.

Table of Contents

1. Introduction

This document serves as the official user manual for the **Student Grade Management System** — a Python-based command-line application designed to streamline the grading workflow for academic instructors. The system supports the complete lifecycle of grade management: from data entry and automated score calculation, through statistical analysis and visualization, to final report export.

This manual guides the instructor through every available screen and menu option — covering how to select a start mode, enter or upload student scores, configure attendance thresholds, view summary statistics and visualizations, and export the final results to CSV files. All instructions are presented from the instructor's perspective to ensure smooth, efficient operation during grading tasks.

2. System Overview

The diagram below illustrates the high-level architecture and data flow of the Student Grade Management System.

■■■ **Teacher (User)**	Initiates the process by providing student data
■ **Data Input**	Manual entry · CSV upload · Import processed summary
■■ **Processing**	Attendance & threshold handling · Weighted total score · Letter grade assignment
■ **Analysis & Reports**	Summary statistics · Score histograms · Grade pie charts · Multi-subject analysis · Student search · CSV export
■ **Output Delivery**	Tables · Charts · Search results · Exported grade files

■ System Design Principle

The system uses a menu-driven, single-instance Python architecture. All data is held in memory during the session and exported to CSV on demand. The attendance threshold mechanism ensures fair, configurable grade penalization per subject.

3. Use Case Overview

The following table summarizes all use cases available to the instructor within the Grade Management System.

Use Case	Actor	Description
Enter Student Data (Manual)	Instructor	Manually input student ID, name, and exam scores subject by subject
Upload Raw CSV	Instructor	Load one or more CSV files containing raw student score data
Import Processed Summary	Instructor	Load an already-processed grade_summary.csv for analysis without re-entry
Set Attendance Threshold	Instructor	Define the maximum absences allowed before attendance penalty is applied
View Summary Statistics	Instructor	Display per-subject and combined count, mean, median, IQR, top/bottom performers
View Score Distribution	Instructor	Plot histograms of TotalScore distributions per subject
View Grade Distribution	Instructor	Display pie charts showing the percentage breakdown of A/B/C/D/F grades
Search Student by Roll Number	Instructor	Look up all subject records for a specific student using their numeric ID
Multi-Subject Analysis	Instructor	Identify students enrolled in multiple subjects and compare cross-subject performance
Export Grade Summaries (CSV)	Instructor	Generate one grade_summary_[Subject].csv file per loaded subject

4. Program Workflow

The diagram below depicts the complete decision flow of the Student Grade Management System from launch to exit. The workflow adapts based on the instructor's starting mode selection.

■	**Launch**	Program starts — Welcome screen displayed with feature overview
2	**Start Mode Selection**	Instructor selects: Option 1 → Import existing processed summary (Analysis Mode) Option 2 → Start fresh with new subject scores (Main Menu) Option 3 → Exit
3	**Analysis Mode (Option 1)**	Load summary CSV → View statistics / charts / search → Return to start or exit
4	**Main Menu (Option 2)**	Enter data manually OR load from CSV(s) → Process scores → View/export results → Loop or exit
5	**Per-Subject Processing**	Set subject name · Total classes · Absence threshold → System calculates: AttendanceScore · WeightedTotalScore · LetterGrade → Store in memory
6	**Post-Processing Options**	View statistics · Plot histograms · Plot pie charts · Search student · Multi-subject analysis · Export CSVs
■	**Exit**	Instructor selects 0 → Goodbye message → Program terminates

5. Step-by-Step User Instructions

STEP

5.1

Starting the Grade Management System

Launch the Python program — Welcome screen appears

When the program is run, a welcome screen is displayed with a brief overview of all system capabilities. The instructor is then prompted to select a start mode.

Terminal Output — Welcome Screen & Start Menu

```
==============================================
STUDENT GRADE MANAGEMENT SYSTEM
==============================================
Features include:
- Manual data entry for student scores
- Loading grade data from CSV files
- Calculating weighted total scores
- Assigning letter grades (A/B/C/D/F)
- Summary statistics and analytics
- Histograms and grade distribution charts
- Exporting final results to CSV
==============================================
How would you like to start?
==============================================
1. Import existing processed grade summary
2. Start fresh and load/process new subject scores
3. Exit.
==============================================
Enter 1, 2 or 3: _
```

STEP **5.2**

Importing an Existing Processed Summary

Option 1 — Load a previously exported grade_summary.csv

If the instructor selects **Option 1**, the system prompts for a processed summary CSV file (e.g., *grade_summary.csv*). After loading, the instructor enters **Analysis Mode** — a read-only exploration environment for already-processed data.

Option	Description
`Option 1`	Show summary statistics (per subject + combined)
`Option 2`	Plot score distribution histogram (TotalScore per subject)
`Option 3`	Plot letter grade pie chart (per subject)
`Option 4`	Search student details by Roll Number
`Option 0`	Return to start menu

STEP **5.3**

Starting Fresh — Main Menu

Option 2 — Enter new subject data manually or via CSV

Selecting **Option 2** opens the **Main Menu**, the primary interface for loading, processing, and analyzing new subject data.

Option	Description
`Option 1`	Enter new student data — manual keyboard input
`Option 2`	Load grade data from one or more CSV files
`Option 3`	Show summary statistics (per subject + combined)
`Option 4`	Plot score distribution histogram
`Option 5`	Plot letter grade pie chart
`Option 6`	Search student details by Roll Number

`Option 7`	Export grade summaries per subject to CSV
`Option 0`	Exit the program

STEP

5.4

Entering Student Data Manually

Main Menu → Option 1 — Input scores via keyboard

The manual entry mode guides the instructor through a structured prompt sequence for each student. The system requests subject configuration first, then collects per-student data until the instructor types **q** to finalize the subject.

Field	Description
`Subject Name`	Enter a descriptive name, e.g., 'Data Science' or 'Machine Learning'
`Total Classes`	The total number of scheduled classes held for this subject
`Absence Threshold`	Maximum absences allowed before an attendance penalty is applied to TotalScore
`Student ID`	Numeric student identifier — type q to stop entering students
`Student Name`	Full name of the student
`Midterm 1 Score`	Score out of 100
`Midterm 2 Score`	Score out of 100
`Final Exam Score`	Score out of 100
`Missed Classes`	Number of classes this student was absent from

Terminal Output — Manual Entry Session

```
Enter your choice: 1
Enter subject name: Business Scripting
Enter total number of classes: 32
Enter allowed absences before penalty: 6

Student ID (numeric) or 'q' to stop: 1001
Student Name: Rahul Pandit
Midterm 1 score (0-100): 88
Midterm 2 score (0-100): 92
Final Exam score (0-100): 95
Number of missed classes: 2
```

```
Student added.

Student ID (numeric) or 'q' to stop: q

Total scores and letter grades have been calculated.
```

■ Automatic Calculations on Entry Completion

Once 'q' is entered, the system automatically computes: AttendanceScore (based on missed classes vs. threshold), WeightedTotalScore (Midterm 1: 30% · Midterm 2: 30% · Final: 40% · Attendance penalty), and LetterGrade (A: 90+ · B: 80-89 · C: 70-79 · D: 60-69 · F: below 60).

STEP 5.5

Loading Student Data from CSV Files

Main Menu → Option 2 — Bulk upload via CSV file(s)

The CSV loading mode is designed for bulk data ingestion. The instructor uploads one or more CSV files, each representing a subject. The system processes each file identically to manual entry — calculating attendance, totals, and grades automatically.

Required CSV Column Format:

Student_ID	Student_Name	Midterm_1	Midterm_2	Final_Exam	Missing_Classes
Numeric ID	Full name	Score 0–100	Score 0–100	Score 0–100	Integer ≥ 0

Terminal Output — Multi-Subject CSV Loading

```
Enter your choice: 2

CSV Loading Mode
Enter CSV filename: ISQS_6335_student_scores.csv
Enter subject name (or Enter to use filename): Data Science
Enter total number of classes: 32
Enter allowed absences before penalty: 6

Total scores and letter grades have been calculated.
CSV loaded successfully as subject 'Data Science'.

Do you want to upload scores of a different subject?
1. Yes 2. No
Enter your choice: 1

Enter CSV filename: ISQS_4370_student_scores.csv
Enter subject name: Machine Learning
Enter total number of classes: 32
Enter allowed absences before penalty: 6

Total scores and letter grades have been calculated.
```

```
CSV loaded successfully as subject 'Machine Learning'.
```

STEP 5.6

Viewing Summary Statistics

Main Menu → Option 3 — Per-subject and combined statistical analysis

The summary statistics view provides a comprehensive numerical breakdown of student performance for each loaded subject, followed by a combined cross-subject summary when multiple subjects are present.

Basic Statistics	Count · Mean · Median · Std Dev · Min · Max · Q1 · Q3 for Midterm 1, Midterm 2, Final Exam, and TotalScore
Additional Metrics	Number of students · Class average · Class median · IQR (TotalScore)
Best & Worst Performers	Highest and lowest scorer overall and per exam component
Letter Grade Distribution	Count of students in each grade tier: A / B / C / D / F
Combined Statistics	All subjects merged — multi-subject enrolled students identified with cross-subject averages

STEP 5.7

Histograms and Grade Distribution Charts

Option 4 — Score histogram · Option 5 — Grade pie chart

Two visualization options are available. If multiple subjects are loaded, separate charts are generated for each subject in sequence.

Score Distribution (Option 4)	**Type:** Histogram Plots the distribution of TotalScore values across all students in a subject. Reveals whether scores are clustered, widely spread, or skewed — useful for identifying grading curve needs or identifying unusually difficult assessments.
Grade Distribution (Option 5)	**Type:** Pie Chart Displays the percentage of students in each letter grade tier (A, B, C, D, F). Provides an immediate visual assessment of overall class difficulty and grade equity.

STEP 5.8

Searching for a Student by Roll Number

Main Menu → Option 6 — Cross-subject student lookup

The search feature allows the instructor to locate any individual student across all loaded subjects using their numeric Student ID. The result displays a complete per-subject record for that student.

Search Result Fields:

Field	Description
`Student_ID`	The student's numeric roll number
`Student_Name`	Full name of the student
`Subject`	Subject name in which this record exists
`Midterm_1 / Midterm_2 / Final_Exam`	Individual exam scores
`AttendanceScore`	Score contribution from attendance (100 if within threshold)
`TotalScore`	Weighted composite score including attendance adjustment
`LetterGrade`	Final letter grade assigned based on TotalScore

■ Multi-Subject Enrollment Detection

If a student appears in more than one loaded subject, all records are displayed sequentially — enabling the instructor to compare that student's performance across courses in a single lookup.

STEP 5.9

Exporting Grade Summaries

Main Menu → Option 7 — Generate per-subject CSV export files

The export function generates one CSV file per loaded subject. These files are suitable for official record-keeping, further analysis in Excel, or archival purposes.

Terminal Output — Export Confirmation

```
Enter your choice: 7

Exporting grade summaries for each subject...

Data exported to 'grade_summary_Business_Scripting.csv'.

Data exported to 'grade_summary_Data_Science.csv'.

Data exported to 'grade_summary_Machine_Learning.csv'.

All subject grade summaries exported successfully!
```

Each exported CSV contains the following columns:

Student_ID	Student_Name	Subject	Midterm_1	Midterm_2	Final_Exam	Missing_Classes	AttendanceScore	TotalScore	LetterGrade

STEP

5.10

Exiting the System

Option 0 at any menu — Graceful program termination

The instructor can exit from any menu level by selecting **Option 0**. The system returns to the start menu if exiting from the main menu, or terminates the program entirely if Option 3 is selected at the start menu.

Terminal Output — Exit Sequence

```
Enter your choice: 0

==============================================
How would you like to start?
==============================================
1. Import existing processed grade summary
2. Start fresh and load/process new subject scores
3. Exit.
==============================================
Enter 1, 2 or 3: 3

Thank you for using the program. Goodbye!
```

6. Reports & Visualizations

The system generates seven distinct output types, providing both numerical and graphical perspectives on student performance to support evidence-based grading and instructional decisions.

■ **Summary Statistics**	Displays mean, median, standard deviation, min/max, IQR, and best/worst performers per subject and in a combined cross-subject view. Enables the instructor to identify performance trends and grade distribution patterns at a glance.
■ **Score Distribution Histogram**	Plots TotalScore values as a histogram for each subject. Reveals clustering, spread, and skewness of the score distribution — useful for identifying whether a grading curve adjustment is needed.
■ **Letter Grade Pie Chart**	Shows the percentage of students in each letter grade tier (A, B, C, D, F) per subject. Provides an immediate visual of overall course difficulty and grade equity across the class.
■ **Multi-Subject Student Report**	Identifies all students enrolled in more than one loaded subject. Displays each student's scores, attendance, TotalScore, and LetterGrade across all subjects, plus per-student average statistics for holistic performance tracking.
■ **Student Search Result**	Returns a complete multi-subject record for any student searched by Roll Number — including all exam scores, attendance, TotalScore, and LetterGrade. Ideal for advising sessions and individual grade inquiries.
■ **CSV Grade Summary Export**	Generates one structured CSV file per subject (grade_summary_[Subject].csv) containing all processed student data. Ready for import into Excel, LMS systems, or institutional grade databases.
■ **Attendance-Adjusted Scoring**	The system calculates AttendanceScore based on an instructor-defined absence threshold. Students exceeding the threshold receive a proportional deduction, ensuring attendance is fairly and transparently factored into the final grade.

7. Program Features Summary

Feature	Detail
Data Entry	Dual-mode input: manual keyboard entry OR CSV file upload
Attendance Handling	Configurable absence threshold per subject — automatic penalty calculation
Score Calculation	Weighted total score: Midterm 1 (30%) + Midterm 2 (30%) + Final Exam (40%) + Attendance adjustment
Letter Grade Assignment	Automatic assignment: A ≥ 90 · B ≥ 80 · C ≥ 70 · D ≥ 60 · F < 60
Summary Statistics	Count, mean, median, std dev, IQR, best/worst performers per subject and combined
Visualizations	TotalScore distribution histograms + letter grade pie charts per subject
Student Search	Cross-subject lookup by numeric Student ID — displays all subject records
Multi-Subject Analysis	Detects students enrolled in multiple subjects, computes cross-subject averages
CSV Export	One structured CSV export file per loaded subject — ready for records or further analysis
Import Mode	Load and analyze a previously exported processed summary without re-entering data

8. Data Structures Used

Structure	Purpose & Usage
`Dictionary`	Stores the subjects collection keyed by subject name — provides O(1) lookup by subject
`List`	Holds per-student record sets during manual entry before DataFrame conversion
`DataFrame (pandas)`	Primary data structure for storing and processing student records — enables vectorized statistical operations

`Set`	Used for identifying students enrolled in multiple subjects (via ID intersection across subject DataFrames)
`Array (NumPy)`	Underlies DataFrame operations — used in histogram and statistical calculations

9. Major Functions

Function	Purpose
`load_csv(filename)`	Loads a raw CSV file into a pandas DataFrame and returns it for processing
`score_input(prompt)`	Validates and returns a numeric score input — enforces 0–100 range with error handling
`export_to_csv(df, filename)`	Exports a processed grade DataFrame to a named CSV file in the working directory
`calculate_total_score_attendance(row)`	Computes AttendanceScore based on absence threshold, then calculates weighted TotalScore for a single student row
`assign_letter_grade(score)`	Maps a numeric TotalScore to a letter grade (A/B/C/D/F) using defined grade boundaries
`add_totals_and_grades(df)`	Applies calculate_total_score_attendance() and assign_letter_grade() to every row in a DataFrame
`show_summary_statistics(df, title)`	Prints detailed statistical summary including basic stats, additional metrics, best/worst performers, and grade distribution
`show_all_statistics(subjects)`	Iterates all loaded subjects, calls show_summary_statistics() per subject, then generates combined cross-subject statistics
`collect_student_data()`	Interactive loop that prompts the instructor for each student's data until 'q' is entered — returns a list of dictionaries
`search_student(roll_number, subjects)`	Searches all loaded subject DataFrames for a given Roll Number and prints all matching records
`handle_manual_entry(subjects, all_data_df)`	Orchestrates the full manual data entry flow — calls collect_student_data(), converts to DataFrame, calculates totals, and stores the subject
`handle_csv_loading(subjects, all_data_df)`	Manages CSV file upload mode — loops allowing multiple subject CSV uploads with per-subject configuration
`handle_recalculate(subjects, all_data_df)`	Recalculates all scores and grades for every loaded subject — useful after threshold adjustments
`handle_plot_scores(subjects)`	Generates and displays TotalScore histograms for each loaded subject using matplotlib

`handle_plot_grades(subjects)`	Generates and displays letter grade pie charts for each loaded subject using matplotlib
`handle_export(subjects)`	Exports processed grade DataFrames to CSV files for all loaded subjects
`handle_import_existing_summary(subjects, all_data_df)`	Imports a pre-existing processed grade summary CSV into the subjects dictionary for Analysis Mode

10. Conclusion

The Student Grade Management System represents a complete, instructor-centered academic data management solution. By combining automated score calculations, configurable attendance thresholds, rich statistical analysis, and clear visual outputs, the system empowers instructors to make informed, evidence-based grading decisions with minimal manual effort.

The system's menu-driven architecture ensures accessibility for users at all technical skill levels while providing the depth of functionality required for managing single and multi-subject grade data. Key strengths include:

- Dual input modes (manual and CSV) accommodate different instructor workflows
- Attendance threshold mechanism ensures transparent, fair grade penalization
- Multi-subject analysis provides a holistic view of student performance across courses
- Statistical outputs and visualizations support data-driven curriculum and assessment decisions
- CSV export ensures seamless integration with institutional record-keeping systems
- Student search by Roll Number enables fast, accurate grade lookups for advising sessions

■ Project Summary

This system was developed as part of ISQS 6335 at Texas Tech University. It demonstrates practical application of Python programming, data structures (dictionaries, lists, DataFrames, sets), statistical computing with pandas/NumPy, and data visualization with matplotlib in a real-world academic management context.

11. Appendix — Sample Statistical Output

The following shows a representative multi-subject statistical summary output from the system, demonstrating the depth of analysis available when two subjects are loaded simultaneously.

Subject 1: Business Scripting (50 students)

Terminal Output — Business Scripting Statistics

```
SUMMARY STATISTICS: Subject — Business Scripting
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
Basic Statistics
count mean std min 50% max
Midterm_1 50 82.80 10.17 62.0 83.5 99.0
Midterm_2 50 82.88 11.60 50.0 86.5 99.0
Final_Exam 50 79.50 14.57 50.0 82.0 99.0
TotalScore 50 82.63 8.06 66.5 83.7 95.7

Number of Students : 50
Class Average : 82.63
Class Median : 83.68
IQR (TotalScore) : 12.66

Highest Scorer : Dana Brown (95.70)
Lowest Scorer : Shawn Baker (66.50)

Letter Grade Distribution
A: 10 B: 24 C: 11 D: 5
```

Subject 2: Machine Learning (51 students)

Terminal Output — Machine Learning Statistics

```
SUMMARY STATISTICS: Subject — Machine Learning
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
Basic Statistics
count mean std min 50% max
```

```
Midterm_1 51 90.29 10.84 70.0 95.0 100.0
Midterm_2 51 92.86 9.08 72.0 97.0 100.0
Final_Exam 51 90.00 9.58 70.0 92.0 100.0
TotalScore 51 91.75 5.38 81.4 91.8 100.0

Number of Students : 51
Class Average : 91.75
Class Median : 91.75
IQR (TotalScore) : 9.38

Highest Scorer : Alex Wilson (100.00)
Lowest Scorer : Jamie Hall (81.35)

Letter Grade Distribution
A: 30 B: 21
```

Combined (All Subjects — 101 records)

Terminal Output — Combined Cross-Subject Statistics

```
SUMMARY STATISTICS: All Subjects (Combined)
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
count mean std min 50% max
Midterm_1 101 86.58 11.12 62.0 88.0 100.0
Midterm_2 101 87.92 11.51 50.0 90.0 100.0
Final_Exam 101 84.80 13.33 50.0 86.0 100.0
TotalScore 101 87.23 8.20 66.5 88.4 100.0

Number of Students : 101
Class Average : 87.23
Class Median : 88.40
IQR (TotalScore) : 9.95

Highest Scorer (All) : Alex Wilson (100.00)
Lowest Scorer (All) : Shawn Baker (66.50)

Letter Grade Distribution (Combined)
```

```
A: 40 B: 45 C: 11 D: 5

Multi-Subject Students : 10 students enrolled in both subjects
```

■ Interpreting the Combined Statistics

The combined view reveals that Machine Learning students performed notably higher on average (91.75 TotalScore) compared to Business Scripting students (82.63 TotalScore). Across both subjects combined, 84% of students earned A or B grades — indicating strong overall class performance. The 10 multi-subject students can be individually tracked to identify whether performance in one course predicts performance in the other.